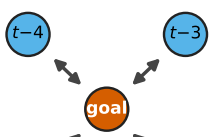
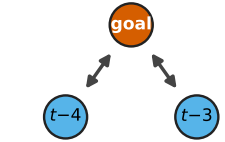
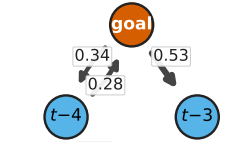
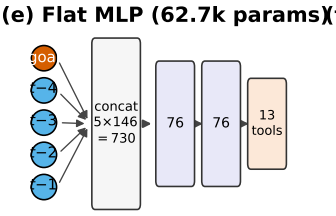
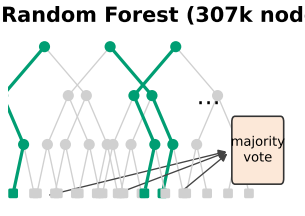

(a) 1D Chain
goal
t−4
t−3
t−2
t−1
Nearest-neighbor line: goal needs
V−1 = 4 steps to reach latest call

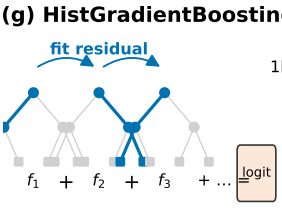

(b) Star GNCA
goal
t−4
t−3
t−2
t−1
Central hub: goal broadcasts to
every cell in exactly 1 step

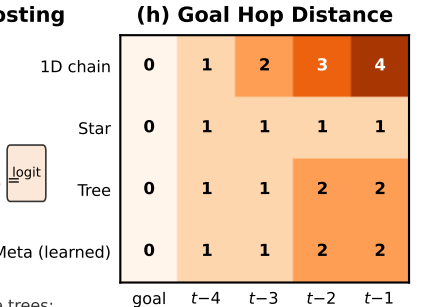

(c) Tree GNCA
goal
t−4
t−3
t−2
t−1
Binary call tree: recent calls are
siblings with O(log V) diameter


(d) Learned Topology
goal
0.34
0.53
0.28
t−4
t−3
0.55
0.53
t−2
t−1
Row-softmax edge weights initialised
at the tree; largest weights, seed 0


(e) Flat MLP (62.7k params)
goal
t−4
t−3
t−2
t−1
concat
5×146
= 730
76
76
13
tools
No graph: all cells concatenated;
cell identity is solely input offset


(f) Random Forest (307k nodes)
majority
vote
Deep bootstrap trees voted in parallel;
each split tests a single feature


(g) HistGradientBoosting
fit residual
f1 + f2 + f3 + ...
logit
Shallow sequential additive trees;
each tree fits prior pseudo-residuals


(h) Goal Hop Distance
1D chain
Star
Tree
Meta (learned)
0
1
2
3
4
goal
t−4
t−3
t−2
t−1
Steps for goal intent to reach cell;
Star & Tree achieve O(1)/O(log V) hops